*framework and tools project*

# SALES ANALYSIS

BY
NORAH ALKABKABI 444003676
AMIRAH MOHSEN ALGHAMWY 444005124
JOORY HANI HALABI 444000428

## INTRODUCTION

**This report summarizes the findings, methodologies, and conclusions from the Data Analysis and Machine Learning Pipeline Development project. The project was divided into three phases: Data Wrangling and Exploration with Pandas, Numerical Computation with NumPy, and Machine Learning with Scikit-learn. The goal was to apply fundamental data analysis and machine learning techniques to solve a real-world problem using a structured dataset.**

## DATASET OVERVIEW

THE DATASET USED IN THIS PROJECT IS THE AVOCADO SALES DATASET, WHICH CONTAINS INFORMATION ABOUT AVOCADO SALES ACROSS DIFFERENT REGIONS IN THE UNITED STATES. KEY FEATURES INCLUDE:

- DATE: THE DATE OF THE OBSERVATION.
- AVERAGEPRICE: THE AVERAGE PRICE OF AVOCADOS.
- TOTAL VOLUME: TOTAL VOLUME OF AVOCADOS SOLD.
- SMALL BAGS, LARGE BAGS, XLARGE BAGS: VOLUME OF AVOCADOS SOLD IN DIFFERENT BAG SIZES.
- TOTAL BAGS: TOTAL VOLUME OF AVOCADOS SOLD IN BAGS.
- TYPE: TYPE OF AVOCADO (CONVENTIONAL OR ORGANIC).
- REGION: THE REGION WHERE THE AVOCADOS WERE SOLD.

# PHASE 1: DATA WRANGLING AND EXPLORATION WITH PANDAS

**1. DATA LOADING AND EXPLORATION**

**THE DATASET WAS LOADED USING PANDAS, AND INITIAL EXPLORATION WAS CONDUCTED TO UNDERSTAND ITS STRUCTURE AND CONTENTS. KEY STEPS INCLUDED:**

- **DISPLAYING THE FIRST FEW ROWS OF THE DATASET.**
- **CHECKING FOR MISSING VALUES, DUPLICATES, AND OUTLIERS.**
- **SUMMARIZING THE DATASET USING DESCRIPTIVE STATISTICS.**

**2. DATA CLEANING**

**THE DATASET WAS RELATIVELY CLEAN, BUT THE FOLLOWING STEPS WERE TAKEN TO ENSURE DATA QUALITY:**

- **HANDLING MISSING VALUES: NO MISSING VALUES WERE FOUND.**
- **REMOVING DUPLICATES: NO DUPLICATE ROWS WERE IDENTIFIED.**
- **OUTLIER DETECTION: NO SIGNIFICANT OUTLIERS WERE DETECTED.**

**3. EXPLORATORY DATA ANALYSIS (EDA)**

**EDA WAS CONDUCTED TO UNCOVER PATTERNS AND RELATIONSHIPS IN THE DATA:**

- **SUMMARY STATISTICS: BASIC STATISTICS (MEAN, MEDIAN, STANDARD DEVIATION) WERE CALCULATED FOR NUMERICAL COLUMNS.**
- **DATA VISUALIZATION: VISUALIZATIONS SUCH AS SCATTER PLOTS, HISTOGRAMS, AND BOXPLOTS WERE CREATED TO EXPLORE THE DISTRIBUTION OF KEY FEATURES LIKE "AVERAGEPRICE" AND "TOTAL BAGS".**
- **INSIGHTS: KEY INSIGHTS INCLUDED THE STRONG CORRELATION BETWEEN "SMALL BAGS" AND "TOTAL BAGS", AS WELL AS SEASONAL TRENDS IN AVOCADO PRICES.**

**4. DATA MANIPULATION**

**PANDAS WAS USED TO MANIPULATE THE DATA:**

- **FILTERING: DATA WAS FILTERED TO FOCUS ON SPECIFIC YEARS OR REGIONS.**
- **TRANSFORMING COLUMNS: NEW FEATURES WERE CREATED, SUCH AS LAG FEATURES FOR TIME SERIES ANALYSIS.**

# PHASE 2: NUMERICAL COMPUTATION WITH NUMPY

1. VECTORIZED OPERATIONS
NUMPY WAS USED TO PERFORM EFFICIENT VECTORIZED OPERATIONS ON THE DATASET:
CALCULATIONS OF MEAN, MEDIAN, AND VARIANCE FOR NUMERICAL COLUMNS.
CREATION OF A CORRELATION MATRIX TO IDENTIFY RELATIONSHIPS BETWEEN FEATURES.
2. CUSTOM MATRIX OPERATIONS
A CUSTOM ALGORITHM WAS IMPLEMENTED TO PERFORM MATRIX OPERATIONS, SUCH AS MATRIX MULTIPLICATION AND TRANSPOSITION, TO SOLIDIFY UNDERSTANDING OF NUMERICAL COMPUTING PRINCIPLES.
3. PERFORMANCE OPTIMIZATION
NUMPY'S VECTORIZED OPERATIONS SIGNIFICANTLY IMPROVED THE PERFORMANCE OF NUMERICAL COMPUTATIONS COMPARED TO TRADITIONAL LOOPS.

# PHASE 3: MACHINE LEARNING WITH SCIKIT-LEARN

1. DATA PREPROCESSING

THE DATASET WAS PREPROCESSED TO PREPARE IT FOR MACHINE LEARNING:
SCALING: NUMERICAL FEATURES WERE SCALED USING STANDARDSCALER.
ENCODING CATEGORICAL VARIABLES: CATEGORICAL FEATURES LIKE "TYPE" AND "REGION" WERE ENCODED USING LABELENCODER.
TRAIN-TEST SPLIT: THE DATASET WAS SPLIT INTO TRAINING AND TESTING SETS.

2. MODEL DEVELOPMENT

TWO MACHINE LEARNING MODELS WERE DEVELOPED:

LINEAR REGRESSION:
A SIMPLE LINEAR REGRESSION MODEL WAS BUILT TO PREDICT "TOTAL BAGS" BASED ON "SMALL BAGS". THE MODEL ACHIEVED AN $R^2$ SCORE OF 0.92.

GRADIENT BOOSTING AND RANDOM FOREST:
MORE ADVANCED MODELS WERE IMPLEMENTED, WITH GRADIENT BOOSTING ACHIEVING THE BEST PERFORMANCE ($R^2$ SCORE OF 0.94).

3. MODEL OPTIMIZATION

TECHNIQUES LIKE GRIDSEARCHCV AND CROSS-VALIDATION WERE USED TO OPTIMIZE HYPERPARAMETERS AND IMPROVE MODEL PERFORMANCE.

4. MODEL EVALUATION

THE MODELS WERE EVALUATED USING METRICS SUCH AS:

MEAN SQUARED ERROR (MSE)
R-SQUARED ($R^2$)

RESIDUAL ANALYSIS:
RESIDUAL PLOTS WERE USED TO CHECK FOR OVERFITTING.

# CHALLENGES FACED

**OVERFITTING: INITIAL MODELS-APPLE SALES- SHOWED SIGNS OF OVERFITTING, SO WE CHAGNED IT TO THE CURRENT DATASET.**

**FEATURE SELECTION: IDENTIFYING THE MOST RELEVANT FEATURES REQUIRED CAREFUL ANALYSIS AND DOMAIN KNOWLEDGE.**

**DATA QUALITY: ENSURING THE DATASET WAS CLEAN AND READY FOR ANALYSIS WAS TIME-CONSUMING BUT ESSENTIAL**